

IoT assignment Q2

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107-13

Ques 1. Smart irrigation system: Arduino UNO and ESP32

→ I have to build a smart irrigation system that reads soil moisture, control a water pump and sends data to a mobile application every 5 minute

The main reason is that ESP32 is much more suitable for IoT application than Arduino UNO. A smart irrigation system is not only sensing moisture and switching a pump ON/OFF but also sending data to a mobile app continuously. For that ESP32 already has build-in WIFI and bluetooth. Arduino UNO does not have this feature by default, so I would have to add an extra WIFI module like ESP8266, which increases wiring, complexity and chances of error.

Ques 2 (a) Would my answer change if internet was not required,

yes. If the system only has to read soil moisture and control a pump locally, Arduino UNO can also do that job. It is cheaper and simpler for a basic local irrigation system.

Ques 2 Smart factory with 500 sensor across 2 km: compare WiFi, Bluetooth, LoRa and MQTT

→ A small factory with 500 sensor spread over 2 km needs a communication system that is scalable, reliable

low power, and suitable for long distance. In my opinion best choice would be a combination of LoRa and MQTT with WiFi used only at gateway.

(*) WiFi

WiFi provides high data rate and easy internet connectivity, but it has limited range.

(*) LoRa

LoRa is highly suitable because it provides long range with low power consumption. It is ideal for sensors that only send small packets such as temperature, vibration, pressure on tank level.

(*) MQTT

MQTT is not a radio technology, it is a light weight messaging protocol. It works very well between gateway and cloud/server.

Ques 3 Sensors automatically show impossible value such as -120°C and 300°C . How to locate the fault?

→ If the sensor suddenly start showing impossible value, I would troubleshoot the system step by step instead of assuming the sensor is damaged.

① Check whether value is physically realistic. Since -120°C and 300°C is impossible.

- ② Inspect the sensor physically
- ③ Replace the sensor with another known working sensor. If the new one works, the old sensor is faulty
- ④ Check wiring and continuity. Look wires, broken jumper connection
- ⑤ Check the power supply. Measure the sensor VCC and verify that the ground is common
- ⑥ Check the ADC path if the sensor is analog
- ⑦ Print raw value on serial monitor. This helps identify whether the value is the sensor signal or in software conversion
- ⑧ Check software formulas and library usage.
- ⑨ Check for noise and interference from motor, relays or long wires

Ques 4 ESP32 weather station should run 6 months on battery. What changes would you make

- If an ESP32 weather station has to run on battery for six months, then the design must focus heavily on power saving. The ESP32 must sleep most of the time and wake up only when needed.

⊕ Hardware changes

- ↳ Use a high capacity battery with a power low quiescent current regulator.
- ↳ Remove unnecessary always on components such as bright LED or unused modules

low

(#) software changes

(a) ↳ use deep sleep mode.

this is very important step

↳ wake up only every few minute instead of continuous running

Ques 5 Automatic delay system needs new delay which communication system would you use?

→ I would choose CAN bus (Controller Area network)

- ① Very low latency, suitable for real time control
- ② Prioritize based communication, so critical brake messages can be transmitted quickly
- ③ Robust against electrical noise found in vehicles
- ④ More reliable than wireless operation such as wifi or Bluetooth
- ⑤ Already widely used in automotive system such as ABS

Q6 Average classroom temperature using only one sensor, where shall it be placed?

To estimate the average classroom temperature using only one sensor, I would place the sensor at the center of the class and at a moderate corresponding roughly to human

If there is an actuator

ESP32 → Relay driver → Pump / Alarm

Possible Failure

- ① Sensor Failure
 - ↳ use sensor validation
- ② Relay Failure
 - ↳ use proper driver circuit and isolation
- ③ Cloud Failure
 - ↳ keep local backup and handle server error
- ④ Loose wiring
 - ↳ use proper connection and soldering

Ques 9 If attacker gets access through wifi, what attacks are possible and how to improve security?

- If an attacker gain access to IoT device through wifi: several attacks are possible
 - ① unauthorized control of relay, pump
 - ② Inject of false sensor data
 - ③ Theft of data, password

way to improve security

- ① use a strong WPA2 / WPA3 wifi password
- ② use encrypted communication
- ③ Change default username and password

~ not seeing new reading while offline

Flow

1.1.1

Ques 10 ESP 32 with DHT 11, ultrasonic sensor, OLED, relay and soil moisture sensor: what conflicts can occur?

→ Possible conflicts include

- ① GPIO conflicts
 - ↳ wrong pin assignment
- ② Voltage mismatch
 - ↳ some module may use 5V while ESP 32 uses 3.3V
- ③ Power instability
 - ↳ soil moisture conflict
- ④ Relay noise
 - ↳ switching load can disturb sensor reading

Solution

- ↳ plan pin mapping carefully
- ↳ use level shifting if needed
- ↳ provide stable power and decoupling capacitors
- ↳ ensure all ground are common

Ques 11 Internet fails for 12 hours. How to ensure no data is lost?

→ The best solution is to use a store and forward strategy

- ① read sensor data every minute as usual
- ② Attach a timestamp to each reading
- ③ If internet is available, upload normally
- ④ Keep storing new reading while offline

⑤ Periodically test the internet connection

⑥ when internet return, upload all stored data first, oldest to newest

Ques 12 Devices must run for five years without human intervention. What changes would you make?

→ For five year continuous operation, I would redesign the system for long term reliability.

Hardware changes

① Use industrial or high quality component instead of fragile hobby modules

② Use a proper PCB instead of breadboard wiring

Software changes

① Use watchdog timer to recover from hang

② Validate sensor reading and reject impossible values

Ques 13 Can an IoT system be smart without AI?

difference between smart and intelligent

→ Yes an IoT system can still be called even if it does not use AI. A system is smart if it can sense, make decision based on rules.

Ex: - A smart irrigation system that turns a pump ON when soil moisture is low and OFF when moisture becomes sufficient. It is clearly a smart system even if it uses pulse counting.

* faster and

Flow

11.5.1

uses only threshold logic.

∞ differences

Smart system

(*) uses sensor and auto
mation

(*) responds according
to predefined rules

(*) does not necessarily
learn

Intelligent system

(*) learn from data
or past behaviour

(*) adapts or predicts

(*) may use AI / ML
techniques

Q114 Polling vs interrupt based programming

→ Polling

In polling, the microcontroller repeatedly checks whether an event has occurred. It is simple to understand and useful for small projects but it uses CPU time and may miss fast events if the loop is slow

→ Interrupt

In interrupt based programming, the event itself notifies the CPU. The processor temporarily pause its normal work.

Comparison

Polling

* Simple but less efficient

* Interrupt accepted
when timing is not
critical

Interrupt

* IOT performance
is increased here
in cases such as
→ pulse counting

* faster and
efficient

Flow

Start

- ↳ initialise ESP32
- ↳ read sensor data
- ↳ add timestamp
- ↳ check wifi

If wifi connected

- try upload
- if upload successful, check if any old stored data exist
- upload stored backlog
- continue

If not connected

- save current reading locally
- wait for next reading cycle
- repeat

Ques 17 Flowchart for ESP32 OTA update with error handling

→ Flow start

- boot ESP32
- connect to wifi
- if wifi fails, retry
- check update server for new firm ware version
- if no update available, continue normal operation
- if update available, download firmware

- write firmware to OTA partition
- if writing fails, abort update and keep old firmware
- reboot ESP32
- if new firmware boots successfully, mark update successful
- if new firmware fails, return back to previous firmware or safe mode if supported

11m ~
am 15 These design trade-offs in an IoT project

- Trade off 1: Cost vs accuracy
 - ↳ Using sensor such as DHT11 or low cost soil moisture probes reduces cost but sacrifices measurement accuracy
- Trade off 2: Power vs frequent update
 - ↳ Keeping wifi on and sending data frequently gives better real time monitoring
- Trade off 3: Simplicity vs Scalability
 - ↳ Writing the project in a single loop delay makes it easier to build and debug

Note

These trade-offs are often justified in student project because the goal is to create a working prototype within limited time

On 16 Flowchart for local storage during wifi failure and upload after recovery

- Programs involved
 - ↳ ESP32 firmware
 - ↳ wifi
 - ↳ HTTP S or MQTT for cloud upload
 - ↳ SD card for storage